CAPSTONE 1 · SYSTEM ANALYSIS AND DESIGN

Library Borrowing System

Book Loan and Return

PROJECT TEAM

Team name: [PLACEHOLDER]

Team members: [PLACEHOLDER]

Instructor: [PLACEHOLDER]

LIBRARY BORROWING SYSTEM

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SYSTEM RATIONALE

The Problem and Proposed Solution

Problem

- Manual or disconnected library records.
- Hard to track available books and overdue loans.
- Errors and delays in borrowing and returning.

Proposed solution

A centralized system for members, books, loans, returns, overdue tracking, copies, reports, and borrowing history.

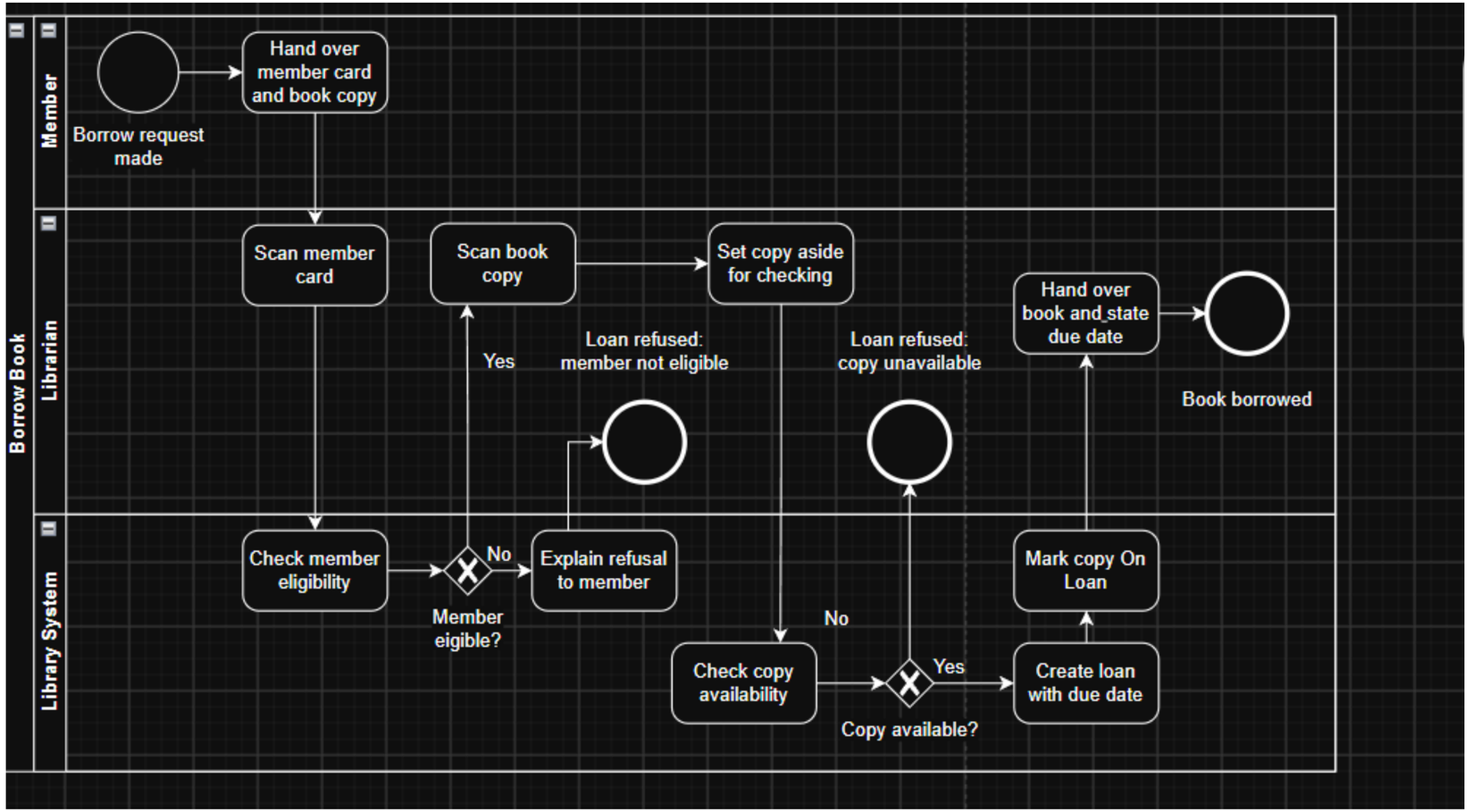
MAIN USERS

Member · Librarian · Library Manager · System Administrator

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Business Process Models (BPMN)

Three process diagrams: BPM1 — Borrow Book · BPM2 — Return Book · BPM3 — Monitor Overdue Loans



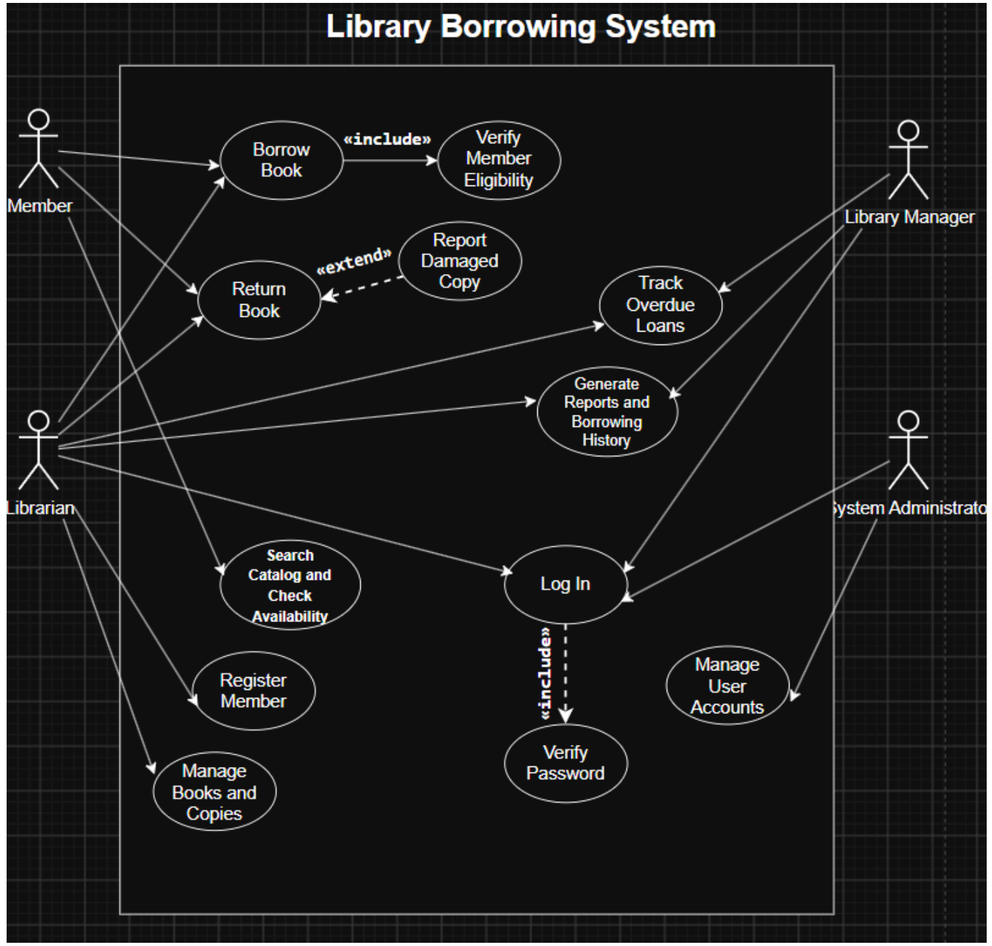
BPM1 focus

1. Member requests a book.
2. System checks eligibility and availability.
3. Librarian scans the physical copy.
4. System creates the loan and assigns a due date.

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UML Use Case Model

One overall Use Case Diagram containing 12 use cases.



ACTORS

Member · Librarian · Library Manager · System Administrator

KEY USE CASES

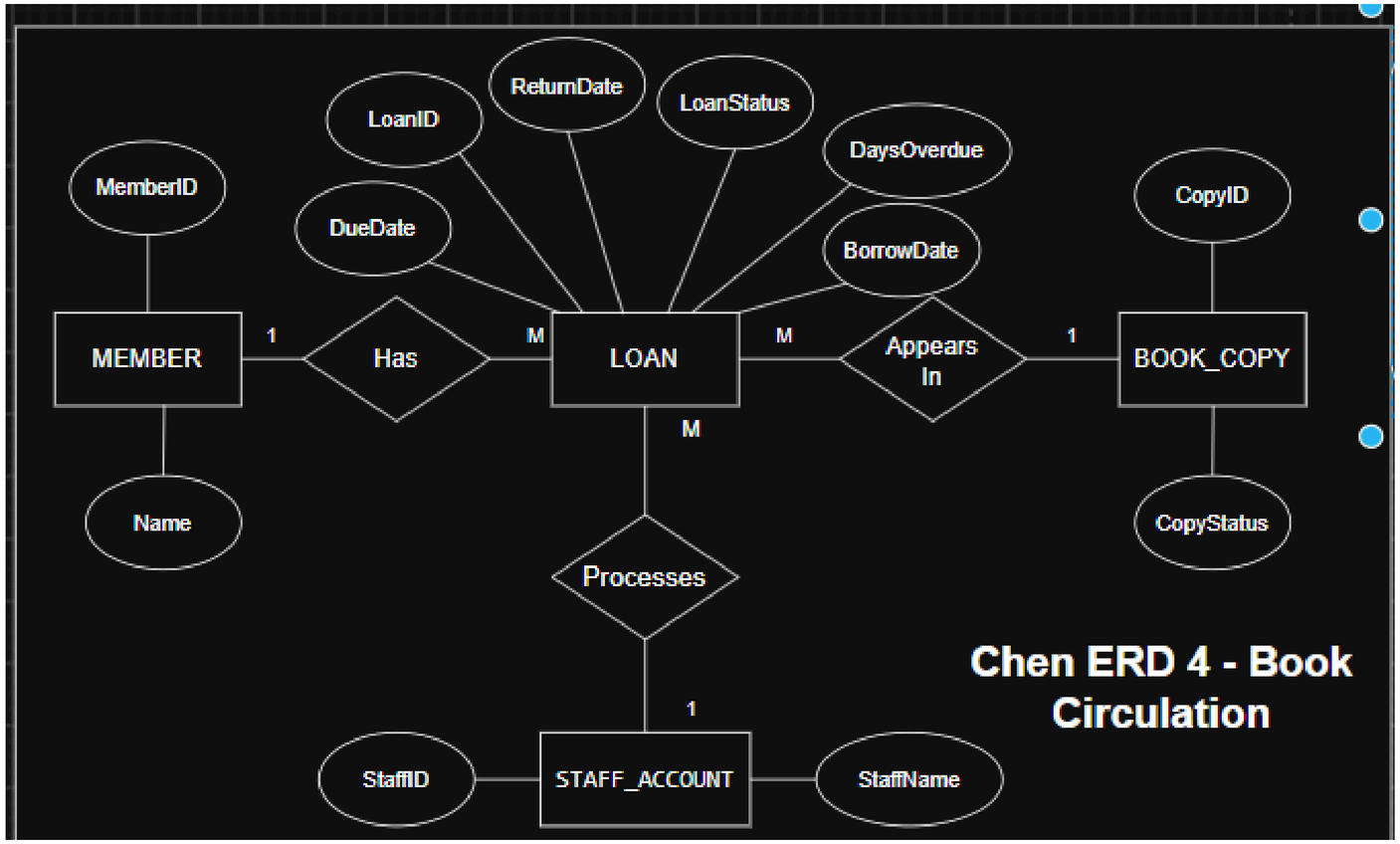
Borrow Book · Return Book · Register Member · Manage Books and Copies · Track Overdue Loans · Generate Reports and Borrowing History

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CONCEPTUAL DATABASE DESIGN

Conceptual Database Design — Chen Notation

Four conceptual ERDs: ERD1 — Book Catalog · ERD2 — Book Inventory · ERD3 — Users · ERD4 — Book Circulation



MAIN ENTITIES

MEMBER · BOOK_COPY · LOAN · STAFF_ACCOUNT

CORE RELATIONSHIPS

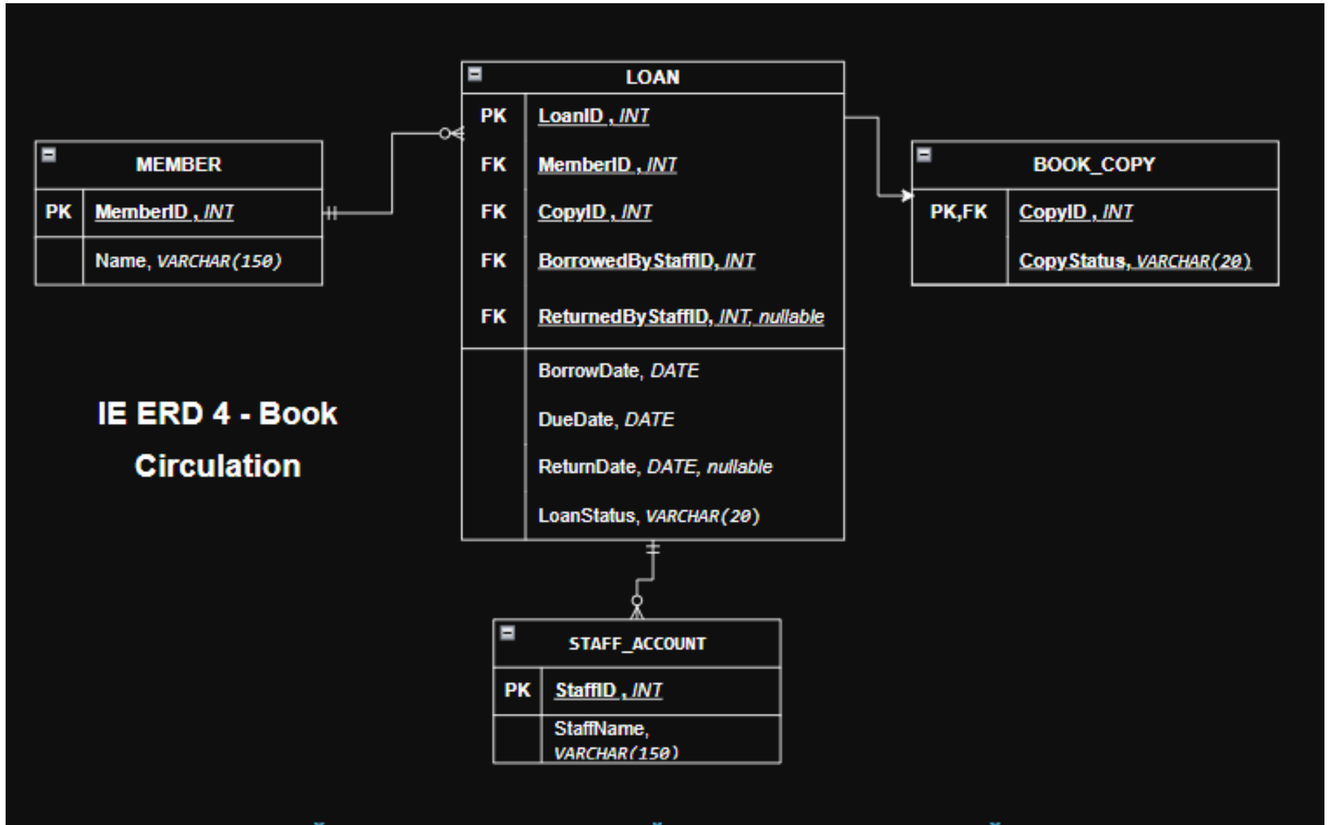
- One member can have multiple loans.
- Each loan refers to one physical book copy.
- Staff accounts are associated with loan processing.

UNDERLINED ATTRIBUTES = IDENTIFIERS · CARDINALITIES = RELATIONSHIP CONSTRAINTS

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Logical Database Design — IE Notation

Four IE ERDs correspond to the four Chen models.



MAIN TABLES

MEMBER · BOOK_COPY · LOAN · STAFF_ACCOUNT

LOAN FOREIGN KEYS

MemberID (FK) · CopyID (FK) · BorrowedByStaffID (FK) · ReturnedByStaffID (FK, optional until return)

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INTEGRATED VIEW

One System — Different Modeling Perspectives

Each model answers a different question about the same Library Borrowing System.

1. PROJECT PLAN AND REQUIREMENTS
2. BPM — How activities happen
3. UML USE CASE — What functions users perform
4. CHEN ERD — What information the system needs
5. IE ERD — How information is stored in tables

EXAMPLE: BORROWING A BOOK

BPM1 describes the process. Borrow Book represents the function. Chen ERD4 identifies the entities. IE ERD4 shows the tables and foreign keys.

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DELIVERABLE

Final Project Report

System Analysis and Design — Mini-Project Final Report

Capstone 1: System Analysis and Design | Instructor: Prof. S. S. Kim

Team Name: Hexagon Group / Section: I25B, I25C, I25D

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Submission date: _____

Contents

Section	Content
1	Project Plan
2	Business Process Models (BPM)
3	UML Use Case Modeling
4	Chen Entity Relationship Diagrams
5	IE Entity Relationship Diagrams
6	Consistency, Methodology and Conclusion

This report presents the five required deliverable areas. Diagram file links are provided in the relevant sections.
The report documents the proposed system design; it does not claim that software has been implemented.

1. Project Plan

1.1 Project background and objectives

Small and medium-sized libraries may use paper records or disconnected tools to manage books, members, loa
and returns. This can cause incorrect availability information, slow transactions and difficulty identifying overdu
loans. The proposed Library Borrowing System centralizes these records and supports library staff in routine
circulation activities.

The project objectives are to document system requirements, model borrowing and return processes, describe
system functions, and develop conceptual and logical data models. The work covers the planning, analysis and
design stages of the systems development life cycle.

Report sections

- Project Plan and Requirements
- Business Process Models
- UML Use Case Modeling
- Chen Entity Relationship Diagrams
- IE Entity Relationship Diagrams

Strengths

Separate sections, clear diagram logic, consistent terminology and business rules, and an explanation of how the models connect.

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CLOSING SUMMARY

Conclusion

Process

Analysis of borrowing, returning, and overdue monitoring.

Function

Identification of system users and their interactions with the system.

Data

Conceptual and logical database design.

Our project provides a structured foundation for developing a Library Borrowing System. We applied BPMN, UML, Chen ERD, and IE ERD methods to analyze processes, define system functions, and organize data.

Thank you for your attention.

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